

Hans Wassermann

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EDUCATION

University of Michigan

Aug 2026 – May 2028

Master of Science in Electrical and Computer Engineering

Ann Arbor, MI

- Control Systems Specialization
- **Relevant Coursework:** Embedded Control Systems, Guidance & Navigation of Aerospace Vehicles

Rutgers University

Sep 2023 – May 2026

Bachelor of Science in Electrical and Computer Engineering | 3 Year Track

New Brunswick, NJ

- **GPA:** 4.00 / 4.00
- Dean's List (All Semesters), Engineering Honors Academy, Learning Assistant, Matthew Leydt Society
- **Relevant Coursework:** Control System Design, Linear Systems and Signals, Digital Signal Processing, Embedded Systems

EXPERIENCE

Undergraduate Research Assistant — Robotics & Controls

May 2026 – Aug 2026

SWARM Intelligence Lab | MATLAB, Python, C, Git, OpenCV, Serial Communication

New Brunswick, NJ

- Developed a **multi-agent motion planning system** in Python to steer 4 magnetic fish across a 75x75 solenoid grid in real time, using camera feedback at 20Hz and a lookahead waypoint buffer to follow trajectories generated by a **physics-based schooling simulation**
- Implemented **inter-agent repulsion dynamics** in the schooling model to enforce a hard minimum spacing constraint across all agents, preventing collisions while preserving emergent collective behavior
- Designed stall detection and recovery logic using **camera-based position feedback**, including waypoint-skip and coil-reanchor strategies validated through log analysis

Augmented Reality Research Intern

May 2025 – Aug 2025

Rutgers WINLAB | C#, Unity, OpenCV, Python, MQTT, GitLab, Linux

New Brunswick, NJ

- Extended a **multi-user collaborative AR** drawing platform by building avatar and username systems, syncing user positions and movement across HoloLens clients in real time using **MQTT**
- Designed a **QR-code-based spatial anchoring system**, integrated into the app's launch procedure, to **synchronize coordinate frames** across multiple AR headsets
- Prototyped a **real-time body-tracking pipeline** using an Intel RealSense depth camera and **YOLO pose estimation**, serializing pose data over MQTT to reconstruct user movement on other clients

Data Engineering Intern

May 2024 – Aug 2024

BASF ECMS | Python, Jupyter Notebook, Excel, VBA

Iselin, NJ

- Automated **real-time data processing for 20+ reactor test protocols in Python**, cutting manual analysis time for engineers evaluating vehicle emission catalyst performance
- Built **automated quality-control checks** into the data pipeline, flagging out-of-range sensor readings before they reached reactor personnel
- Developed an **interactive dashboard** for engineers to access and visualize raw and processed test data
- Scaled the **automation framework globally** to BASF R&D sites in China, Germany, and other facilities

PROJECTS

Beehive Monitoring System | C++, Python, PlatformIO, FastAPI, PyTorch, vLLM, Onshape

Sep 2025 – May 2026

- Designed an **embedded, real-time sensing system** on a dual-ESP32 architecture with a **power-constrained, duty-cycled control loop**, balancing sampling frequency against battery life for **autonomous field deployment**
- Fine-tuned a **transformer-based audio classifier (AST)** on beehive audio to detect queen presence, then integrated a context-aware **LLM chatbot via a custom MCP server** for natural-language querying of live sensor data
- Built a **full-stack telemetry platform** (FastAPI backend, REST API, web dashboard) to ingest and visualize real-time and historical sensor data from embedded hardware

Neuron Model Control System Design | MATLAB, Simulink, Nonlinear Control, Automated Testing

Sep 2025 – Dec 2025

- Linearized the nonlinear Hindmarsh-Rose neuron model about analytically computed equilibria, performed **Jacobian-based stability analysis** to classify fixed points, and validated spiking and bursting dynamics against theoretical outputs in MATLAB/Simulink
- Designed and tuned **state-feedback and PID controllers** to regulate neuronal firing rate, using root locus and frequency-domain methods to meet **closed-loop performance** specifications

TECHNICAL SKILLS

Programming: Python, MATLAB, C, C++, C#, Java, Verilog

Hardware & Test: Oscilloscope, Function Generator, Multimeter, Power Supply, Soldering, LTspice, KiCAD

Tools: Git, Simulink, Linux, Onshape, Arduino IDE, Jupyter Notebook, Unity, Microsoft Office